

Yunjae Won

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EDUCATION

KAIST Graduate School of AI

Integrated M.S.+Ph.D., Advisor: Minjoon Seo and Seongjoon Oh

Republic of Korea

Fall 2024 – Present

KAIST

Bachelor of Science in Electrical Engineering, Minor in Computer Science

Republic of Korea

Feb. 2018 – Aug. 2024

GPA: 3.76/4.3

PUBLICATION

HyperCLOVA X 32B Think

NAVER Cloud HyperCLOVA X Team

Technical Report

Instruction Tuning with and without Context: Behavioral Shifts and Downstream Impact

EACL 2026 Oral

Hyunji Lee, Seunghyun Yoon, Yunjae Won, Hanseok Oh, Geewook Kim, Trung Bui, Franck Dernoncourt, Elias Stengel-Eskin, Mohit Bansal, Minjoon Seo

Differential Information Distribution: A Bayesian Perspective on Direct Preference Optimization

Preprint (Under Review)

Yunjae Won, Hyunji Lee, Hyeonbin Hwang, Minjoon Seo

EXPERIENCE

Foundation Model Research Internship

NAVER Cloud HyperCLOVA Pre-Training Team

Republic of Korea

Aug. 2025 – Nov. 2025

- Built a scalable reasoning-data filtering pipeline using a 0.6B reward model, replacing resource-heavy >80B LLM-based filtering and improving KMMLU accuracy by 4.72%p.
- Led cleansing for 450K expert Q&A samples; removed 99.87% of detected sensitive PII information and doubled average answer length by improving response completeness.
- Played a key role in improving MMLU-Pro accuracy by 8.82%p by augmenting Q&A options, generating reasoning traces, and filtering low-quality synthetic data.

Research Internship

NAVER G Place AI Development

Republic of Korea

Mar. 2023 – Aug. 2023

- Led the implementation of an end-to-end workflow using black-box optimization techniques to compare data augmentation, preprocessing, and hyperparameter choices at a distributed scale.
- Using this pipeline, identified a silent data processing error for training VQA models; fixing the issue reduced model size by over 10× while improving accuracy by 56%; model has been in service since July 2023.

Undergraduate Research Assistant

Data Intelligence Lab, KAIST School of Electrical Engineering

Republic of Korea

Jan. 2023 – Feb. 2023

- Studied contrastive-learning approaches for active learning and evaluated how common active learning methods degrade under instance-dependent and class-dependent label noise on real-world datasets.

Research Internship

NCSOFT Speech AI Lab

Republic of Korea

July 2022 – Aug. 2022

- Led development of an in-the-wild data collection pipeline for training singing source separation models; eliminated a signal-processing bottleneck by reformulating the computation as a convex optimization problem and using its closed-form solution, reducing complexity from $\mathcal{O}(n)$ to $\mathcal{O}(1)$.
- Increased training dataset size by 12x and led to a new state-of-the-art model, improving Signal-to-Distortion Ratio from 8.06 to 8.45; received the Excellent NCSOFT Summer Internship Project award.

PROJECTS

Generating Harmful Responses from Aligned Models

Spring 2025

- Theoretically analyzed the optimal sampling distribution of rejected responses for Direct Preference Optimization, and demonstrated that one can generate *harmful* responses using safety-aligned policies without any re-training or sophisticated prompt-injection attacks.

Orthogonal Gradient Descent: Learning from Preferences with Minimal Forgetting

Fall 2024

- Developed a projected gradient descent method that prevents likelihood decrease on chosen responses during preference alignment, reducing forgetting by up to 99.15%.

Metabolic Reaction Prediction via Next Token Prediction

Fall 2024

- Formulated various metabolic reaction prediction tasks as a single next token prediction problem, which allowed training one transformer architecture that can handle multiple tasks.
- Proposed approach achieved SOTA accuracy, outperforming other task-specific, graph neural network based methods.

Korean Text Recognition Challenge |

Jan. 2023

- Ranked 1st out of 1,158 participants in the Kyowon Group AI Challenge of developing a Korean children handwriting recognition model.
- Outperformed an OCR corporate team backed by high-end GPUs, using only free-tier Google Colaboratory resources.

RESEARCH INTERESTS

- Preference Optimization, Reinforcement Learning from Human Feedback
- Knowledge Distillation, On-Policy Distillation
- Continual Learning
- Constrained Optimization